

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR.IIT_*CO-SC(MODEL-A& APEX) **Date:18-12-24**
Time: 3HRS **GTM-1** **Max. Marks: 300**

Name of the Student: _____ **H.T. NO:** | | | | | | | |

18-12-24_SR.* CO-SUPER CHAINA (MODEL-A& APEX)_JEE-MAIN_GTM-1_SYLLABUS

PHYSICS: Waves, Ray Optics And Optical Instruments, Wave Optics, Electrostatics, Capacitors, Current Electricity, Magnetism, EMI & AC, EM Waves, Modern Physics

CHEMISTRY: Atomic structure, Liquid solutions, Electrochemistry, Chemical kinetics, d - and f- block Elements, Group-15 to 18, Coordination Compounds, Alcohols, Phenols and Ethers, Aldehyde and Ketones, Carboxylic Acids, Amine, Diazonium Salts, Biomolecules, Carbohydrates, Proteins, Vitamins, Nucleic acids

MATHEMATICS: Complex Numbers, Quadratic Expressions, Sequence & Series, Permutations & Combinations, Binomial Theorem, Probability, Circle, Parabola, Ellipse, Hyperbola, Integration, Differential Equations.

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PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Two waves are approaching towards each other with a velocity of 16m/s and frequency n. The distance between two consecutive nodes is

A) $\frac{16}{n}$ B) $\frac{8}{n}$ C) $\frac{n}{16}$ D) $\frac{n}{8}$

2. Two monoatomic ideal gases 1 and 2 of molecular masses m_1 and m_2 respectively are enclosed in separate containers kept at the same temperature. The ratio of the speed of sound in gas 1 to that in gas 2 is given by

A) $\sqrt{\frac{m_1}{m_2}}$ B) $\sqrt{\frac{m_2}{m_1}}$ C) $\frac{m_1}{m_2}$ D) $\frac{m_2}{m_1}$

3. Three capacitors each of capacitance C and of breakdown voltage V are joined in series. The capacitance and breakdown voltage of the combination will be

A) $\frac{C}{3},\frac{V}{3}$ B) $3C,\frac{V}{3}$ C) $\frac{C}{3},3V$ D) $3C,3V$

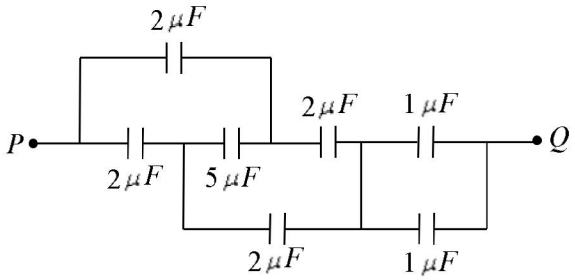
4. The dimension of $\frac{1}{2}\varepsilon_0 E^2$ (ε_0 : permittivity of free space; E : electric field) is

A) MLT^{-1} B) ML^2T^{-2} C) $ML^{-1}T^{-2}$ D) ML^2T^{-1}

5. The electrostatic potential energy between proton and electron separated by a distance $1\mathring{A}$ is

A) $-13.6eV$ B) $27.2eV$ C) $-14.4eV$ D) $-3.41eV$

6. The effective capacitance between the points P and Q of the arrangement shown in the figure is



A) $\frac{1}{2}\mu F$ B) $1\mu F$ C) $2\mu F$ D) $1.33\mu F$

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7. An electric bulb is designed to draw power P_0 at voltage V_0. If the voltage is V it draws a power P. Then

A) $P=\left(\frac{V_0}{V}\right)^2 P_0$ B) $P=\left(\frac{V}{V_0}\right)^2 P_0$ C) $P=\left(\frac{V}{V_0}\right)P_0$ D) $P=\left(\frac{V_0}{V}\right)P_0$

8. When a battery connected across a resistor of 16 Ω, the voltage across the resistor is 12 V. When the same battery is connected across a resistor of 10 Ω, voltage across it is 11V. The internal resistance of the battery (in ohm) is

A) $\frac{10}{7}$ B) $\frac{20}{7}$ C) $\frac{25}{7}$ D) $\frac{30}{7}$

9. A rigid circular loop of radius r and mass m lies in the $x-y$ plane of a flat table and has a current i flowing in it. Consider z axis to be vertically upwards. At this particular place the earth's magnetic field is $\vec{B}=B_x\hat{i}+B_z\hat{k}$. The value of i so that the loop start tilting is (assume friction to be sufficiently

A) $\frac{mg}{\pi r\sqrt{B_x^2+B_z^2}}$ B) $\frac{mg}{\pi r B_x}$ C) $\frac{mg}{\pi r B_z}$ D) $\frac{mg}{\pi r\sqrt{B_x B_z}}$

10. A horizontal overhead powerline is at a height of 4 m from the ground and carries a current of 100 A from east to west. The magnetic field directly below it on the ground due to it is $\left(\mu_0=4\pi\times10^{-7}\ \text{TmA}^{-1}\right)$

A) 2.5×10^{-7} T, southward B) 5.0×10^{-6} T, northward

C) 5.0×10^{-6} T, southward D) 2.5×10^{-7} T, northward

11. If the first order minima in Young's double slit experiment occurs direct by in front of one of the slits. If distance between slits is d and the distance between slits and screen is D, then the wavelength of used light is

A) $\frac{d^2}{D}$ B) $\frac{d^2}{2D}$ C) $\frac{d^2}{3D}$ D) $\frac{d^2}{4D}$

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12. A uniform electric field exists between two opposite charged parallel plates. An electron is released from rest from the surface of the negatively charged plate. The electron strikes the surface of the opposite plate 2.0×10^{-2} m from the first, in 1.5×10^{-8} s. Then, potential difference between plates is

A) 20.2 V B) 2.02 V C) 40.4 V D) 4.04 V

13. A light bulb is at a depth of D below the surface of water. An opaque disc of radius R is placed on the surface of water just above the bulb. The bulb is not at all seen through the surface of water, then (n = Refractive index of water)

A) $R=\frac{D}{\sqrt{n^2-1}}$ B) $R>\frac{D}{\sqrt{n^2-1}}$ C) $R<\frac{D}{\sqrt{n^2-1}}$ D) $R=D\sqrt{n^2-1}$

14. A lens is placed between a source of light and a wall. It forms images of area A_1 and A_2 on the wall for its two different positions. The area of the source or light is

A) $\frac{A_1+A_2}{2}$ B) $\left[\frac{1}{A_1}+\frac{1}{A_2}\right]^{-1}$ C) $\sqrt{A_1A_2}$ D) $\left[\frac{\sqrt{A_1}+\sqrt{A_2}}{2}\right]^2$

15. What should be the velocity of an electron so that its momentum becomes equal to that of a photon of wavelength 5200 $\mathring{A}$?

A) $10^3\,ms^{-1}$ B) $1.2\times10^3\,ms^{-1}$ C) $1.4\times10^3\,ms^{-1}$ D) $2.8\times10^3\,ms^{-1}$

16. A beam of cathode rays is first accelerated to a potential difference V an then is subjected to crossed Electric (E) and Magnetic field (B). The fields are adjusted such that the beam is not deflected. The specific charge of the cathode rays is given by.

A) $\frac{E^2}{2VB^2}$ B) $\frac{B^2}{2VE^2}$ C) $\frac{2VB^2}{E^2}$ D) $\frac{2VE^2}{B^2}$

17. Using the following data

Mass hydrogen atom = 1.00783 u

Mass of neutron = 1.00867 u

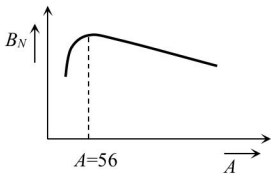
Mass of nitrogen atom ($_7N^{14}$)= 14.00307 u

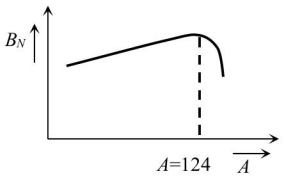
The calculated value of the binding energy of the nucleus of the nitrogen atom ($_7N^{14}$) is close to

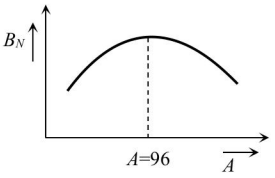
A) 56 MeV B) 98 MeV C) 104 MeV D) 142 MeV

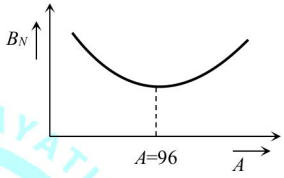
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18. The dependence of binding energy per nucleon, B_N on the mass number, A, is represented by

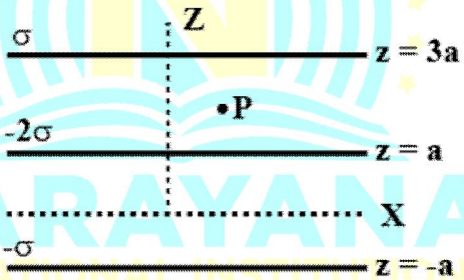
A) 

B) 

C) 

D) 

19. Three infinitely long charge sheets are placed as shown in figure below. The electric field at point P is



A) $\frac{2\sigma}{\varepsilon_0}\hat{k}$ B) $-\frac{2\sigma}{\varepsilon_0}\hat{k}$ C) $\frac{4\sigma}{\varepsilon_0}\hat{k}$ D) $-\frac{4\sigma}{\varepsilon_0}\hat{k}$

20. A solid non-conducting sphere of radius R has a uniform distribution of charge working space throughout its volume. At a distance x from its centre, for $x < R$, the electric field is directly proportional to

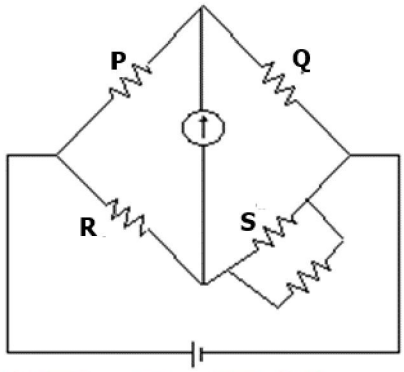
A) $\frac{1}{x^2}$ B) $\frac{1}{x}$ C) x D) x^2

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only.** **Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

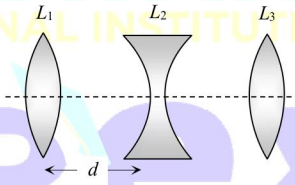
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21. In a Wheatstone's network $P = 2\Omega$, $Q = 2\Omega$, $R = 2\Omega$ and $S = 3\Omega$. The resistance with which S is to be shunted in order that the bridge may be balanced is



22. A uniformly charged solid sphere made of an insulating material has potential 100V at its surface. What is the potential of center of the sphere in Volts?

23. Three lenses L_1, L_2, L_3 are placed co-axially as shown in figure. Focal lengths of lenses are given $30\,cm, 10\,cm$ and $5\,cm$ respectively. If a parallel beam of light falling on lens L_1, emerging L_3 as a convergent beam such that it converges at the second focus of L_3. Distance between L_1 and L_2 will be



24. In a compound microscope, the focal length of the objective and the eye lens are $2.5\,cm$ and $5\,cm$ respectively. An object is placed at $3.75\ cm$ before the objective and image is formed at the least distance of distinct vision, then find the distance between two lenses upto nearest integer in centimeters.

25. A circular coil of 100 turns and effective diameter 20 cm carries a current of 0.5 A. It is to be turned in a magnetic field B = 2 T from a position of its stable equilibrium to unstable equilibrium. The work required in the process to nearest integer in Joules is ____.

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CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

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Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. An electron beam can undergo diffraction by crystals which proves the wave nature of electrons. The potential required for a beam of electrons to be accelerated from rest so that its wavelength becomes equal to 0.154 nm is

A) 63.5 V B) 31.75 V C) 635 V D) 127 V

27. Which of the following azeotropic solutions has the boiling point more than boiling point of the constituents A and B ?

A) CH_3CH_2OH and CH_3COCH_3 B) CS_2 and CH_3COCH_3

C) $CHCl_3$ and CH_3COCH_3 D) CH_3CHO and CS_2

28. If the standard electrode potentials of Fe^{3+}/Fe and Fe^{2+}/Fe are –0.04 V and –0.44 V respectively then that of Fe^{3+}/Fe^{2+} is

A) 0.76 V B) –0.76 V C) 0.40 V D) –0.40 V

29. A current of 0.1 A was passed for 965 second through a solution of Cu^+ solution and 0.03175 g of copper was deposited on the cathode. Calculate the current efficiency for the copper deposition. (Cu – 63.5)

A) 79% B) 39.5% C) 63.25% D) 50%

30. The specific conductance of a N/10 KCl at 25°C is 0.0112 $ohm^{-1}cm^{-1}$. The resistance of cell containing solution at the same temperature was found to be 55 ohms. The cell constant will be

A) $6.16 cm^{-1}$ B) $0.616 cm^{-1}$

C) $0.0616 cm^{-1}$ D) $616 cm^{-1}$

31. The mechanism of the reaction, $2NO + O_2 \longrightarrow 2NO_2$ is

$NO + NO \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} N_2O_2$ (fast); $N_2O_2 + O_2 \xrightarrow{k_2} 2NO_2$ (slow);

The rate constant of the reaction is

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A) k_2 B) $k_2k_1(k_{-1})$ C) k_2k_1 D) $k_2\left(\frac{k_1}{k_{-1}}\right)$

32. $KMnO_4$ in excess on treatment with concentrated H_2SO_4 forms a compound (X) which decomposes explosively on heating forming (Y). The (X) and (Y) are respectively are

A) Mn_2O_7, MnO_2 B) Mn_2O_7, Mn_2O_3

C) $MnSO_4, Mn_2O_3$ D) Mn_2O_3, MnO_2

33. MnO_4^{1-} is of intense pink colour, though Mn is in +VII oxidation state. This is due to ___

A) d-d transition

B) charge transfer when Mn gives electron to oxygen

C) charge transfer when oxygen gives its electrons to Mn making it +VI

D) absorption of IR frequencies

34. **Consider the following statements**

S1 : $\left[Cr(NH_3)_6\right]^{3+}$ is n inner orbital complex with crystal field stabilization energy equal to $-1.2\ \Delta_0$

S2 : The octahedral complex formed by joining the CN^- ligands to Fe^{3+} ion has theoretical value of 'spin only' magnetic moment equal to 1.73 B.M.

S3 : $Na_2S + Na_2\left[Fe(CN)_5NO\right] \longrightarrow Na_4\left[Fe(CN)_5NOS\right]$, In reactant and product the oxidation states of iron are same and arrange in the order of True/False.

A) F T F B) T T F C) T T T D) F F F

35. The **CORRECT** order of stability of oxides of various halogens is

A) Cl > Br > I B) Br < Cl < I C) Cl < Br < I D) Br > Cl > I

36. There is considerable increase in covalent radius from N to P. However, from Sb to Bi only small increase (of 7 pm) in covalent radius is observed. This is due to

A) poor shielding by completely filled d- and f-orbitals in Bi.

B) similar electronegativity of Sb and Bi

C) Bi being the last element of the group.

D) similar densities of Sb and Bi.

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37. Which of the following acids is a vitamin?

A) Aspartic acid B) Ascorbic acid C) Adipic acid D) Saccharic acid

38. Select the **CORRECT** option among the following

A) n-Butylamine has higher boiling point than n-Butyl alcohol

B) triethyl amine is more basic than ethyl amine in aqueous solution

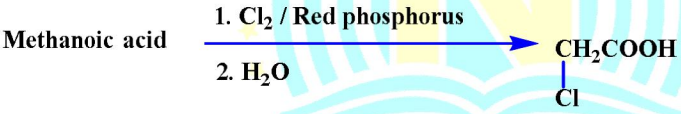
C) 2-methylpropan-2-amine is an example of tertiary amine

D) All

39. Select the **CORRECT** option from the given below

A) Sodium salt of propionic acid on decarboxylation with sodalime gives propane as a major product

B) Benzoic acid is more acidic than phenyl acetic acid

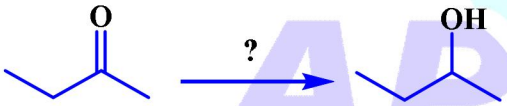
C) 

D) All

40. What is **NOT** true for both cellulose and DNA?

A) Both are long chain polymers B) Both contain similar monomers

C) Both have glycosidic Linkages D) All

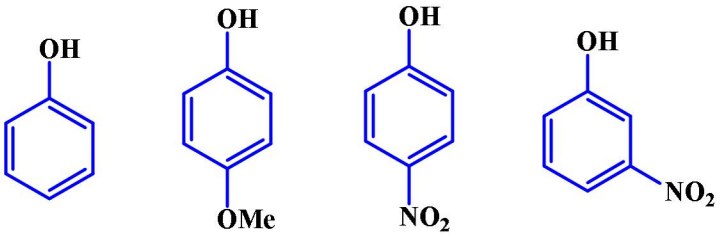
41. 

Which of the following reagents is **NOT** suitable for the above transformation?

A) $NaBH_4$ / H^+ B) $LiAlH_4$ / H^+

C) H_2;Pd / C D) CH_3MgBr / H^+

42. The Ka of



I II III IV

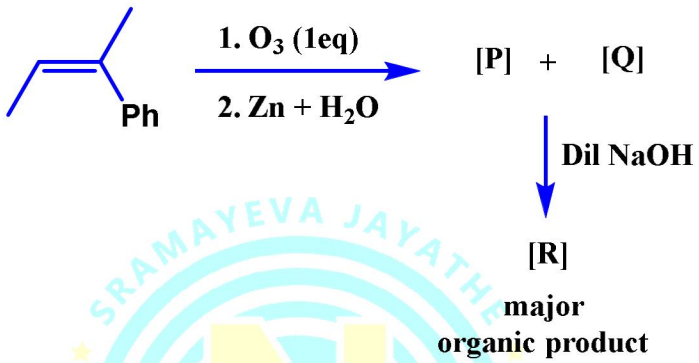
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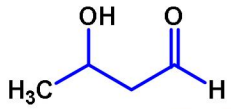
Follows the order

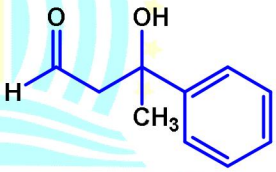
A) III > IV > I > II B) IV > III > II > I

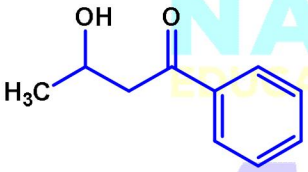
C) II > I > IV > III D) III > II > IV > I

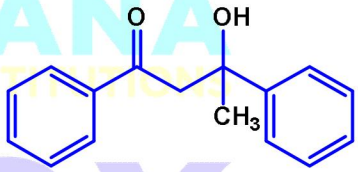
43. Identify the major organic product [R] in the given reaction sequence



A) 

B) 

C) 

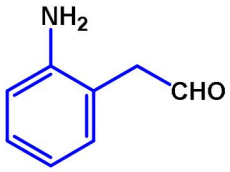
D) 

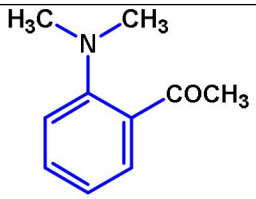
44. The tests performed on compound X and their inferences are:

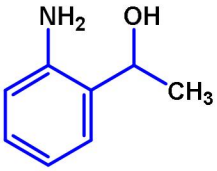
	Test	Interference
(a)	2, 4-DNP test	No coloured precipitate
(b)	Iodoform test	Yellow precipitate
(c)	Azo-dye test	Dye formation

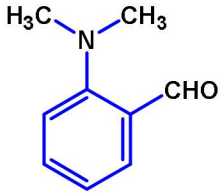
Compound 'X' is

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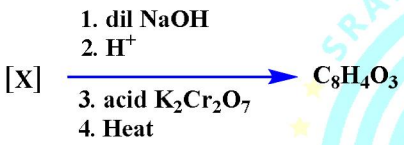
A) 

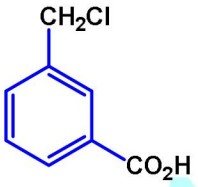
B) 

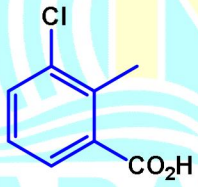
C) 

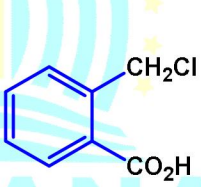
D) 

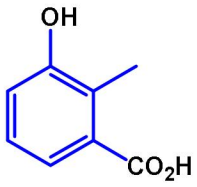
45. The compound 'X' undergoing the following reaction is



A) 

B) 

C) 

D) 

SECTION-II
(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

46. Depression in freezing point of 0.01 molal aq. CH_3COOH solution is 0.02046 °C. 1 molal urea solution freezes at –1.86°C. Assuming molality equal to molarity, pH of CH_3COOH solution is ___

47. Decomposition of H_2O_2 is a first order reaction. A '16' V solution of H_2O_2 of half-life period 30 minutes is present at start. After how many minutes the solution will become '1' V?

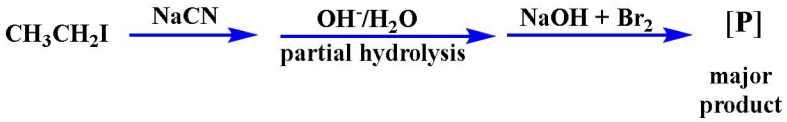
48. How many total isomers (structural + stereo isomers) are possible for a compound with formula, $\left[Rh(en)_2Cl(NO_2)\right]$

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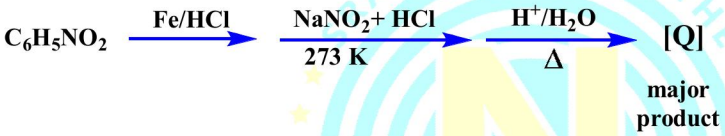
49. Number of different organic products obtained when a mixture of optically pure D-glucose and D-fructose are reduced with excess of $NaBH_4$ in same reaction vessel?

50. Among the given reactions in how many reactions number of carbon atoms is same in starting material and final product (note: consider major organic product in each case)

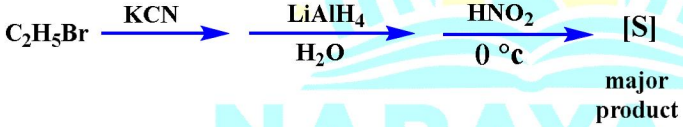
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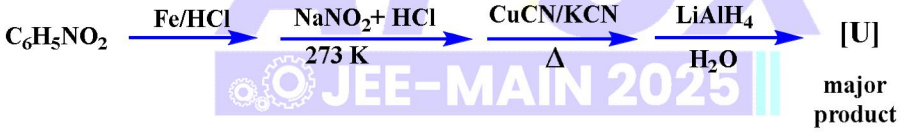
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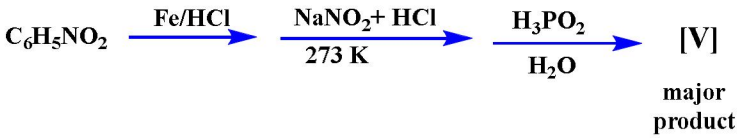
4.



5.



6.



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MATHEMATICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. Let complex numbers z_1, z_2 satisfy $\mathrm{Re}(z_1) > 0$, $\mathrm{Re}(z_2) > 0$ and $\mathrm{Re}(z_1^2) = \mathrm{Re}(z_2^2) = 2$ (Re(z) represents the real part of a complex number z). then the minimum value of $|z_1 + 2| + |\overline{z_2} + 2| - |\overline{z_1} - z_2|$ is _____

A) $\sqrt{2}$ B) $3\sqrt{2}$ C) $4\sqrt{2}$ D) $5\sqrt{2}$

52. The total number of 5-digit numbers, formed by using the digits 1, 2, 3, 5, 6, 7 without repetition which are multiple of 6, is

A) 36 B) 48 C) 60 D) 72

53. Suppose complex numbers z, w satisfy $|z| = 3$ and $(z + \overline{w})(\overline{z} - w) = 7 + 4i$ where 'i' is the imaginary unit, and $\overline{z}$, $\overline{w}$ are conjugates of complex numbers z, w respectively. Then the modulus of $(z + 2\overline{w})(\overline{z} - 2w)$ is

A) $\sqrt{65}$ B) $\sqrt{75}$ C) 65 D) 49

54. In the Cartesian plane, ellipse 'C' is given by $\frac{x^2}{9} + \frac{y^2}{10} = 1$ Point F is the upper focus of C, A is the right vertex C, and P is a moving point on 'C' in the first quadrant. The maximum area of quadrilateral OAPF is

A) $3\sqrt{11}$ B) $\frac{3\sqrt{11}}{2}$ C) $\sqrt{11}$ D) $\frac{\sqrt{11}}{2}$

55. The area of the region described by $A = \{(x, y) : x^2 + y^2 \le 1 \text{ and } y^2 \le 1 - x\}$ is

A) $\frac{\pi}{2} - \frac{4}{3}$ B) $\frac{\pi}{2} - \frac{2}{3}$ C) $\frac{\pi}{2} + \frac{2}{3}$ D) $\frac{\pi}{2} + \frac{4}{3}$

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56. The solution of the differential equation $x.\frac{dy}{dx}+2y=x^2\left(x\neq 0\right)$ with $y(1)=1$, is

A) $y=\frac{x^2}{4}+\frac{3}{4x^2}$ B) $y=\frac{3}{4}x^2+\frac{1}{4x^2}$ C) $y=\frac{4}{5}x^3+\frac{1}{5x^2}$ D) $y=\frac{x^3}{5}+\frac{1}{5x^2}$

57. If $\int\frac{x^2\left(x^6+x^5-1\right)dx}{\left(2x^6+3x^5+2\right)^2}=f\left(x\right)+c$ then $f\left(x\right)=$ _____

(where c is integration constant)

A) $\frac{1}{6}\left(\frac{1}{x^{-3}}+\frac{3}{x^{-2}}\right)^{\frac{1}{2}}$ B) $\frac{1}{2}\left(1+x^{-2}+x^{-5}\right)^{-2}$

C) $-\frac{1}{6}\left(2x^3+3x^2+2x^{-3}\right)^{-1}$ D) $x\left(\frac{x^3}{25}+\frac{x^2}{20}+\frac{1}{10}\right)^{\frac{1}{2}}$

58. Let $A=\left\{\theta\in\left(0,2\pi\right):\frac{1+2i\sin\theta}{1-i\sin\theta}\text{is purely imaginary}\right\}$. Then the sum of the elements in A is

A) 2π B) 4π C) π D) 3π

59. If the numbers appeared on the two throws of a fair six faced die are α and β, then the probability that $x^2+\alpha x+\beta>0$, for all $x\in R$, is

A) $\frac{17}{36}$ B) $\frac{4}{9}$ C) $\frac{1}{2}$ D) $\frac{19}{36}$

60. If α and β are the roots of the equation, $7x^2-3x-2=0$, then the value of $\frac{\alpha}{1-\alpha^2}+\frac{\beta}{1-\beta^2}$ is equal to :

A) $\frac{27}{32}$ B) $\frac{1}{24}$ C) $\frac{3}{8}$ D) $\frac{27}{16}$

61. The 8th common term of the series

$S_1=3+7+11+15+19+.....,S_2=1+6+11+16+21+\ \ldots\ldots$ is _____.

A) 151 B) 171 C) 160 D) 161

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62. The value of $\int_{-\pi}^{\pi} \frac{\cos^2 x}{1+a^x} dx, a > 0$

A) $\pi/2$ B) $a\pi$

C) 2π D) π/a

63. The remainder, when 7^{103} is divided by 17 is _____.

A) 5 B) 12 C) 13 D) 6

64. The maximum value of the term independent of t in the expansion of

$\left(tx^{1/5} + \frac{(1-x)^{1/10}}{t} \right)^{10}$, where $x \in (0,1)$ is

A) $\frac{2.10!}{3(5!)^2}$ B) $\frac{10!}{\sqrt{3}(5!)^2}$ C) $\frac{2.10!}{3\sqrt{3}(5!)^2}$ D) $\frac{10!}{3(5!)^2}$

65. Let $A = \{x_1, x_2, ..., x_7\}$ and $B = \{y_1, y_2, y_3\}$ be two sets containing seven and three distinct elements respectively. Then the total number of functions $f : A \to B$ that are onto, if there exist exactly three elements x in A such that $f(x) = y_2$, is equal to

A) $14.{}^7C_2$ B) $16.{}^7C_3$ C) $12.{}^7C_2$ D) $14.{}^7C_3$

66. Let the solution curve $y = y(x)$ of the differential equation

$\frac{dy}{dx} - \frac{3x^5 \tan^{-1}(x^3)}{(1+x^6)^{3/2}} y = 2x \exp\left\{ \frac{x^3 - \tan^{-1} x^3}{\sqrt{(1+x^6)}} \right\}$

(here exp represents exponential function)

Pass through the origin. Then y(1) is equal to

A) $\exp\left\{\frac{4+\pi}{4\sqrt{2}}\right\}$ B) $\exp\left\{\frac{4-\pi}{4\sqrt{2}}\right\}$ C) $\exp\left\{\frac{1-\pi}{4\sqrt{2}}\right\}$ D) $\exp\left\{\frac{\pi-4}{4\sqrt{2}}\right\}$

67. Four distinct points $(2k,3k), (1,0), (0,1)$ and $(0,0)$ lie on a circle for k equal to :

A) $\frac{2}{13}$ B) $\frac{3}{13}$ C) $\frac{5}{13}$ D) $\frac{1}{13}$

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68. The length of the chord of the parabola $x^2 = 4y$ having equation $x - \sqrt{2}y + 4\sqrt{2} = 0$ is

A) $3\sqrt{2}$ B) $2\sqrt{11}$ C) $8\sqrt{2}$ D) $6\sqrt{3}$

69. Let $P\left(\frac{2\sqrt{3}}{\sqrt{7}}, \frac{6}{\sqrt{7}}\right), Q, R$ and S be four points on the ellipse $9x^2 + 4y^2 = 36$ Let PQ and RS be mutually perpendicular and pass through the origin. If $\frac{1}{(PQ)^2} + \frac{1}{(RS)^2} = \frac{p}{q}$, where p and q are coprime, then p + q is equal to

A) 157 B) 143 C) 137 D) 147

70. $\int x e^{\text{Tan}\,x}\left(2 + x\sec^2 x\right)dx =$

A) $2e^{\text{Tan}\,x} + C$ B) $x^2 e^{\text{Tan}\,x} + C$

C) $\text{Tan}\,x + xe^{\tan x} + C$ D) $x^2 + xe^{\text{Tan}\,x} + C$

SECTION-II

(NUMERICAL VALUE ANSWER TYPE)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals , **Round off to the nearest Integer only. Have to Answer all 5 questions** and question will be evaluated according to the following marking scheme:
Marking Scheme: +4 for correct answer, 0 if not attempted and –1 if not correct.

71. Let 'm' be the real solution to the equation

$$29\left(\frac{17 - 7x}{x + 2} + \frac{8x + 55}{x + 3}\right) = 31\left(\frac{24 - 5x}{x + 1} + \frac{5 - 6x}{x + 4}\right) + 370$$

There are positive integers a, & b such that $m = -\frac{a}{b}$ then the minimum value of $a + b$ is

_____

72. Let $I = \int \frac{(e^x - 1)(\sin x - \cos x) + x\cos x}{\sin^2 x + (e^x - 1 - x)^2} dx = \text{Tan}^{-1}\left(f(x)\right) + C$ where 'C' is the constant of integration and $\lim_{x \to 0} f(x) = 0$, if $\lim_{x \to 0} \frac{f(x)}{x} = \frac{p}{q}$ where $p, q \in N$ then find the least value of p + q

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73. Let N denote the sum of the numbers obtained when two dice are rolled. If the probability that $2^N < N!$ is $\frac{m}{n}$, where m and n are coprime, then $4m-3n$ is equal to

____

74. How many ordered pair of integers (a, b) satisfy all the following inequalities $a^2+b^2<16$, $a^2+b^2<8a$, $a^2+b^2<8b$?

75. Let A_n be the sum of the first n terms of the geometric series $704+\frac{704}{2}+\frac{704}{4}+\frac{704}{8}+.....$ *and* B_n be the sum of the first terms of the geometric series. $1984-\frac{1984}{2}+\frac{1984}{4}-\frac{1984}{8}+.....$ *If* $A_n=B_n$. Then find the value of n $(where\ n\in N)$.

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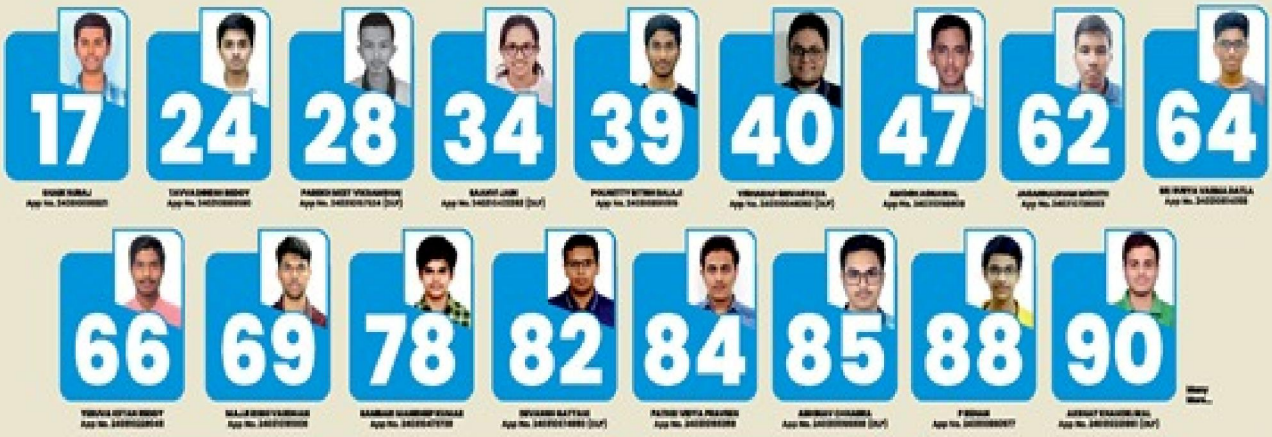
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BHOGALAPALLI SANDESH
HT No: 246138003



AIR 11th

MATCHA BALAADITYA
HT No: 246001057



AIR 12th

RAGHAV SHARMA (DLP)
HT No: 242053073



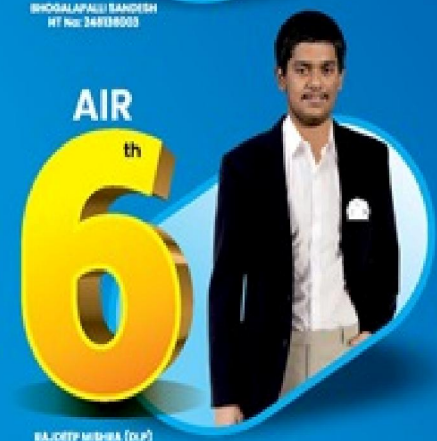
AIR 16th

BISMIT SAHOO (DLP)
HT No: 242072080



AIR 17th

ARYAN PRAKASH
HT No: 241104228



AIR 6th

RAJDEEP MISHRA (DLP)
HT No: 241018170

DREAMERS WHO MADE IT HAPPEN



AIR 20th

AMOGH AGRAWAL
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All India Open Category Ranks in Top 100


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AIR 28


AIR 31


AIR 33


AIR 40


AIR 45


AIR 49


AIR 51


AIR 54


AIR 59


AIR 63


AIR 66


AIR 70


AIR 74


AIR 76


AIR 77


AIR 82


AIR 85


AIR 87


AIR 89


AIR 90


AIR 91


AIR 94


AIR 99

HIGHLIGHTS

Below 20 All India Open Category Ranks	7 Ranks
Below 100 All India Open Category Ranks	31 Ranks
Below 500 All India Open Category Ranks	96 Ranks
Below 1000 All India Open Category Ranks	179 Ranks
Below 100 All India Category Ranks	127 Ranks
Below	873 Ranks

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